

The changes have been saved.



There's WHAT in My Water

How can we make our water safe?

Grade

9-12

Subjects

Science

Estimated Duration

14-19 instructional
hours

Key Content

chemistry, chemical
reactions, chemical
properties, scientific
research,



experimental design,
environmental and
human impact

Students research harmful chemicals in their community water (or the water of a nearby community), identify how people and the environment are affected by these chemicals, and learn how the hazards came to be present. Students examine the chemical pathways and bonding properties of the contaminant, determine the chemical reactions that can effectively neutralize the contaminant, and create an informational product—supported by a scientific report—alerting the community to the dangers/impact of the contaminated water, along with recommended solutions to remediate the issue. ([View a related project video](https://my.pblworks.org/resource/video/water_quality_project) (https://my.pblworks.org/resource/video/water_quality_project).)

Project Path

Project Launch

How safe is our water?

Entry event: Watch the TED-Ed video “When Is Water Safe to Drink?” and participate in an inductive learning experience to learn about the problems caused by contaminants in water.

Form teams and prepare for effective collaboration.

Build Knowledge and Develop & Critique

What happens when water is contaminated?

Learn how water interacts with contaminants; research and write a scientific report on the chemical properties and impact of the team’s selected contaminant.

What is the best way to purify our water so that it is safe?

Conduct experiments to test the efficacy of various contaminant removal/ mitigation methods & individually write a short scientific report summarizing findings and best method of removing/mitigating the selected contaminant.

How can we communicate our findings in order to promote change?

Create a product that informs the community about the problem of water contamination and explains the recommended solution to remediate the issue.

Write team scientific reports to synthesize the key findings and data that support the recommended solution.

Present Products

How can we share our recommendations with the community?

Host a public forum to present informative messages to the community explaining the issue and their recommended solution for addressing the contaminated water problem.

Make It Yours

Reflect on your students

- How can students identify health patterns in their own communities/families to ascertain connections between health and the environment in which they live?
- What might students already know about environmental hazards and environmental justice? How can you explore this content with them in a way that will empower them to effect change in their communities?

Reflect on your context

- What environmental hazards/issues are prevalent in your local community? Who is most impacted by these hazards, and how might you engage them as partners in this project?
- Are there local people and/or organizations already taking action on some of these hazards? How might you partner with them?
- Are there content experts (e.g., chemists, university students or professors, hazardous material

professionals, environmental justice organizers) who could support student teams as they explore the content in this project?

- What conflicting interests (e.g., industry/community members, landlords/tenants) might arise during the course of this project? How will you help students learn about and explore multiple perspectives related to these conflicts?

Reflect on the content & skills

- How can you support students in the generation of the final product? What teachers can you collaborate with in your building to support skills in writing, video, and/or graphic design (dependent on the need if you are partnering with an environmental organization)?
- What opportunities does this project offer to have students engage in experimental design? How will you structure and scaffold this process for your students within the context of the project?
- How will you scaffold students' understanding of the chemistry content (i.e., understanding and usage of periodic table, single displacement reactions, chemical reactions).

Take It Further

Here are some ideas for expanding this project if you choose to do so:

- Be sure to select an environmental issue and contaminants that are relevant to the students' lives and the local geographic context. For example, if soil contamination is a problem in your community, consider having students study soil rather than water. However, if your local community has uncontaminated soil and clean water, seek to identify a nearby community with water-quality issues for students to research and take action on the neighboring community's behalf, in partnership with advocates from the community.
- Guide students to reflect on the source of the contamination and the political/economic context surrounding the presence of these chemicals in a community—including

understanding who is most affected by these hazards and how the hazards came to be present. Explore environmental justice resources such as the following:

- What You Should Know About Environmental Racism
(<https://www.teenvogue.com/story/what-you-should-know-about-environmental-racism>) (Teen Vogue)
- Environmental Justice, Explained (https://www.youtube.com/watch?v=dREtXUij6_c)
(Grist)
- Talking and Teaching Environmental Justice
(<https://www.learningtogive.org/news/talking-and-teaching-environmental-justice>)
(Learning to Give)
- Consider partnering with a local or regional environmental stewardship organization and asking a representative to speak with students about the issue. Coordinate with the organization so that students are making their final products in support of the organization's broader work to address the issue.
- As an alternate entry event, select one of the following options:
 - Have students meet with one or more experts
(<https://my.pblworks.org/resource/working-outside-experts>)/guest speakers (in person or via video conference) to learn about the issue. Possibilities include:
 - Community members whose health or livelihood is being affected by the contaminated water
 - Chemists
 - Representatives from an environmental stewardship organization (see "Notes")
 - Health care professionals
 - University scholars
 - Hazardous material professionals
 - Water quality engineers
 - Have students hear from people affected by the problem via these sources:
 - Videos (news footage, documentaries, etc.)
 - Articles (first person accounts, interviews, etc.)
 - Podcasts
 - Local news media
- Take students on a field trip to a local water treatment plant.

- Engage students in close reading of local water-quality reports and other data that reveal the problem.
- For an additional challenge, have students conduct research to identify specific contaminants and/or understand the relevant water regulations.
- If students have prior knowledge of acids, bases, pH, and acid/base reactions, introduce them to titration as one possible method of contaminant removal and have them determine whether it applies to their contaminant. Demonstrate an acid-base titration lab (<https://www.coursehero.com/study-guides/introchem/acid-base-titrations/>) to determine the concentration of a known acid or base to neutralize the unknown acid or base contaminant. Have students work in teams to practice using the acid-base titration process to neutralize the contaminant. Ask students to determine whether titration is an appropriate method for removing their team's selected contaminant.
- If possible, engage students in fieldwork to test the purity of the local water. Have students take samples from potentially contaminated sources of water in the local community. Support teams as they work together to test those samples for impurities and describe their findings in terms of purity/contamination.
- As an additional challenge in the contaminant removal/mitigation experiments, give students the steps to each procedure on individual cards for them to sort and sequence before testing the procedure on their water sample.
- To support students with their data analysis, consider connecting them with experts in statistics (university students, math teachers, and other professionals who use statistics in their work) and having students present their experiments, hypothesis, and questions about statistical analysis to discuss with the experts.
- For an additional student-driven element, consider having students design and conduct their own experiments to further test specific methods for effectively removing/mitigating water contamination. Help students determine how they will collect and analyze the data to determine the most effective process for producing safe water. Support student inquiry with questions such as these:
 - To what extent is your water sample pure?
 - How could you revise your method of decontaminating your water to get better results?
- Have students report their findings in a lab report describing the purification method they used and explaining their analysis of the results.
- Have students consider whether they will recommend a point-of-use treatment, remediation at the point of origin, and/or a comprehensive strategy to treat the whole system. Encourage students to consider trade-offs and long-term versus short-term

factors when deciding exactly where and how they will recommend their proposed solution be implemented.

- Have students learn about decontamination methods for the other forms of water contamination: physical (filtration or reverse osmosis), biological (distillation or sterilization), and/or radiological (e.g., ultraviolet treatment).

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